Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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Abstract

Under Basel III, banks report Value-at-Risk and Expected Shortfall from a fitted return distribution, and this parametric benchmark is hard to beat on most assets on most days. We develop a semiparametric alternative: a GARCH scale, a residual-quantile model pooled across assets, an extreme-value tail, and a conformal calibration layer. The residual-quantile model is amortized, so a single fit serves hundreds of names and delivers day-one forecasts for newly listed assets. Alongside these forecasts we construct a real-time score of residual-shape stress that flags when the flexible model improves on the parametric standard. The method lowers quantile loss relative to standard benchmarks, with exception counts near nominal at both regulatory levels through the 2008 crisis. The advantage is not uniform; it concentrates in the states the score identifies. One component of that score, fixed in advance, reproduces the concentration out of sample on an untouched 2000–2013 window. The fuller composite we deploy is tuned in sample, so we read its holdout behavior as corroboration rather than as an independent test. We document these results on individual U.S. equities, and across foreign-exchange, country-index, and cross-asset universes.

Keywords: Value-at-Risk; Expected Shortfall; FRTB; quantile regression; extreme value theory; conformal prediction; misspecification; amortized estimation.

JEL classification: C14; C22; C52; C58; G17; G28.

Introduction

Banks measure the downside risk of a trading book by its Value-at-Risk (VaR) and Expected Shortfall (ES): the lower-tail quantile of the conditional return distribution, and the average loss beyond it. Two research traditions produce these forecasts, and they make opposite modeling bets. Econometrics builds parametric filters, such as GARCH and its leverage and skew

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extensions, whose conditional scale dynamics are excellent but whose innovation law is fixed by assumption. Machine learning builds distribution-free conditional quantile estimators, such as gradient-boosted quantile trees, neural implicit quantile networks (IQN), and generative Bayesian computation (GBC) (Polson and Sokolov, 2023), which assume nothing about the innovation law but must learn it from data. Industry practice sits between the two. Under the Basel Fundamental Review of the Trading Book (FRTB), the reported measure is Expected Shortfall at the 97.5% level, the workhorse internal models are historical simulation (HS) and its GARCH-filtered variant (FHS), and forecasts are judged by VaR exception counts at 99% and 97.5%. A method that claims to beat the industry standard must therefore beat HS and FHS on $\mathrm{ES}_{97.5}$ under exception-test discipline. Beating a vanilla GARCH on average loss is not enough.

The advantage of a flexible model over this parametric benchmark is not spread evenly across days, and that unevenness organizes the paper. On the roughly ninety percent of asset-days when residual-shape stress is low, the estimator we deploy and the parametric benchmark are statistically indistinguishable on forecast accuracy, with no detectable penalty for holding the flexible model; Section 4 reports the intervals. The gains are concentrated in the minority of days when residual-shape stress is high. Locating that minority in advance, from an observable signal, is the paper's central task. We construct such a signal, a real-time misspecification score, and show that it orders the flexible model's advantage. We then build a risk model that exploits the ordering while matching the benchmark elsewhere. The generative members of the family go further: they deliver forecasts for assets that have no return history at all, which no parametric point-forecasting model can produce.

Recent-residual tail diagnostics, the excess kurtosis and asymmetry of the GARCH-standardized residuals, predict when distribution-free quantile methods improve on parametric VaR/ES. Where residual-shape stress is high the flexible estimator has a large and significant advantage; where it is low the advantage is statistically indistinguishable from zero in the design panel, a small positive in the point-in-time universe, with no detectable loss. The score's content is the ordering of magnitudes (the top decile is roughly seven times the calm region), not a hard threshold.

The paper contributes to three literatures. To the evaluation of VaR and ES forecasts and the comparison of competing conditional models, it offers the misspecification frontier: a live, real-time, per-asset, per-day score that predicts when distribution-free quantiles beat a GARCH-t benchmark, with per-date Diebold–Mariano significance and validation across four universes: CRSP US equities (top-decile edge +2.71%, DM 6.54), foreign exchange (hyperinflation tier, pooled DM 4.12), twenty-six country indices (a developed-to-frontier gradient), and forty-three cross-asset instruments, with Korea-2026 as a falsification case that sharpens the claim: price crashes are not residual misspecification. To global, pooled forecasting and cross-asset transfer, it shows that a single amortized shape model trained across hundreds of names carries transferable information about conditional tail shape: one fit replaces per-asset estimation of the innovation shape, transfers to names never seen in training, and, in its characteristics-only form, prices assets with no return history at all, the day-one regime in which no per-asset model exists. To semiparametric conditional tail-risk forecasting, it develops the estimator that captures the frontier, a GARCH conditional scale times a state-conditioned nonparametric residual quantile with an extreme-value left tail and an optional split-conformal overlay, which is co-best with CAViaR (Engle and Manganelli, 2004) in the 90% Model Confidence Set (Hansen et al., 2011), improves significantly on GARCH-t, FHS, EWMA, HS, and GJR-skew-t, and attains the lowest joint $(\mathrm{VaR}, \mathrm{ES})$ FZ0 score at both FRTB levels with exception counts near nominal through a 2008-crisis window. Two formal results motivate the search: a pinball regret identity, by which a forecaster's excess loss is locally quadratic in its quantile wedge so an edge concentrates where the wedge is largest, and a tail-wedge proposition, by which innovation-shape misspecification opens that wedge in the tail.

This is a self-contained point-forecasting paper; posterior uncertainty bands, likelihood-free amortized posteriors, and the multivariate portfolio tail are outside its scope.

The loss profile is asymmetric in the score: when the frontier is quiet the estimator ties or edges the parametric benchmark (calm-quintile edge ≈ 0, never negative), and the top-decile improvement is +2.7% (and 12–14% on the hyperinflation currencies where the parametric assumption breaks entirely). These are reductions in a proper scoring rule, not a monetary welfare measure. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level; the joint FZ0 score of Section 5 carries the economic comparison, and no capital-release arithmetic is attempted.

The paper proceeds as follows. Section 1 places the method against the industry benchmark and the forecasting literature; Section 2 develops the semiparametric estimator and the two results that motivate the frontier; Section 3 sets out the data and the forecast-evaluation protocol; Section 4 presents the misspecification frontier; Section 5 reports the joint VaR and ES evaluation; Section 6 covers cross-sectional transfer, longer horizons, and cross-asset evidence; and Section 7 concludes.

1 Related literature and benchmark models

1.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail) and the return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$; the regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 2.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at $\mathrm{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket) while backtesting remains VaR-exception based: more than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint: accuracy on $\mathrm{ES}_{97.5}$ and pinball, and unconditional (Kupiec, 1995) plus independence (Christoffersen, 1998) exception tests at both levels, with significance assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011).

1.2 The parametric family, the simulation family, and CAViaR

Our battery spans the models banks actually run: historical simulation; EWMA/RiskMetrics (Zumbach, 2007); GARCH(1,1)-t (Bollerslev, 1986); GJR-GARCH-skew-t (adding leverage and skewness; Glosten et al., 1993); GARCH-FHS (parametric scale, empirical standardized residuals, the volatile-regime best practice; Barone-Adesi et al., 1999); and SAV-CAViaR (Engle and Manganelli, 2004), the strongest non-nested semiparametric rival among the classical benchmarks. The SAV (symmetric absolute value) specification models the quantile directly on the pinball loss, $q_t(\tau) = \beta_0 + \beta_1\, q_{t-1}(\tau) + \beta_2\, |r_{t-1}|$, the semiparametric analogue of a GARCH recursion written on the quantile.

1.3 Distribution-free conditional quantiles and GBC

The closest antecedent in this journal's own pages is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names, evidence of a shared volatility mechanism that our amortized design also exploits. The differences delimit this paper's contribution. Their target is a point forecast of realized variance; ours is the full conditional return distribution and its regulatory tail, held to calibration, exception-test, and capital discipline. Their question is whether machine learning wins on average; ours is when, and our answer, that on roughly ninety percent of asset-days it does not, reads as a caution against extrapolating average ML superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at every listing age. And we reach the opposite estimator verdict on tabular state (trees over networks), a divergence we attribute to the different target and feature set rather than a contradiction.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (Koenker and Bassett, 1978): gradient-boosted quantile trees (GBM) and the implicit quantile network of Dabney et al. (2018) (IQN: the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). Casting GBC as ERM conditional-quantile regression on observed pairs removes its forward-simulator requirement and extends it to observational time series. The deployed shape model is the trees; the IQN is retained as a benchmark, and its generative reading (draw $\tau \sim U(0,1)$, output $Q_\theta(\tau \mid x)$) is a capability we note without using, since sampling and posterior uncertainty are outside this paper's scope. Generalized Bayes enters through the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\}\pi(\theta)$, a posterior built from a loss in place of a likelihood, targeting the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016); distribution-free finite-sample (exchangeability-based) guarantees enter through split-conformal and adaptive conformal inference (recalibrating predicted intervals on held-out data so that coverage holds without a parametric distributional model, under exchangeability) (Gibbs and Candès, 2021).

2 Semiparametric VaR and ES model

Every estimator below can be reimplemented from the formulas and algorithm boxes given; a glossary of acronyms is in the Online Appendix.

2.1 Notation and objects

- r_t: return of an asset on day t; $\mathcal{F}_{t-1}$: information available before t.
- $Q_t(\tau)$: the conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$, formally $Q_t(\tau) = \inf\{q : \Pr(r_t \leq q \mid \mathcal{F}_{t-1}) \geq \tau\}$: the number the return falls below with probability τ. One sign convention is used throughout: the paper works in *return space*, $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1}\int_0^\tau Q_t(u)\,du \leq q_\tau$, the convention of the FZ scoring function $(v, e < 0)$ and of every table (ES values such as -6.37 are return-space conditional means). The regulatory *loss-space* numbers are their negatives, $\mathrm{VaR}_\tau^{\mathrm{loss}} = -q_\tau$ and $\mathrm{ES}_\tau^{\mathrm{loss}} = -e_\tau$, used only when quoting FRTB levels; the two are never mixed in one expression.
- $z_t = (r_t - \mu_t)/\sigma_t$: the *standardized residual*: the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law.
- s_t: the state vector (trailing realized volatility, misspecification scores, lags) on which nonparametric quantiles condition.
- $\tau \sim \lambda$: a quantile level drawn from a base distribution λ; $H_\phi(s, \tau)$: a learned generator mapping (state, level) to a quantile, in the sense of Polson and Sokolov (2023): $z \overset{D}{=} H(s, \tau)$, $\tau \sim U(0, 1)$, is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u < 0\}), \qquad u = z - q, \tag{1}$$

whose expectation is uniquely minimized by the true conditional quantile (Koenker and Bassett, 1978), the way squared error elicits the mean. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below). The loss is density-free, which is what permits every estimator below to dispense with a likelihood.

Why this loss and not another. We score pinball at the decision-relevant levels rather than a global proper score, for two reasons. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\,d\tau$ (Gneiting and Raftery, 2007); scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. Downside risk lives at fixed regulatory levels ($\tau \in$

$\{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail, and CRPS would average those differences away against a body where all competitors agree. We therefore score pinball *at the levels that are decision-relevant*, and use tail-weighted τ-sampling in training for the same reason. (b) Expected Shortfall is not elicitable on its own (no loss function is minimized by the true ES alone), but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016); the FZ-scored re-evaluation of the battery is a pre-committed extension (Section 7). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them.

2.2 The final model: the residual-hybrid estimator

The deployable model is a four-stage composition. The parametric filter estimates conditional scale, which it does well and with few parameters, and the nonparametric stage estimates the state-dependent shape of the innovation law, which the parametric family fixes by assumption. The extreme-value tail extrapolates beyond the estimation range using the generalized Pareto approximation of Section 2, and the conformal stage corrects the level of the quantile curve using its measured miss rate on a held-out split, with the exchangeability-based guarantee of Section 2.2.

Motivation for the design. Conditional scale is identified by a few parameters and estimated with low variance by a parametric filter, so the shape stage is where flexible methods can add value; per-asset tail data is scarce, which motivates pooling the standardized residuals across names. The EVT stage extrapolates the tail by the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), and the conformal stage is motivated by the split-conformal guarantee of Proposition 1, exact under exchangeability and approximate for the deployed time-ordered residuals. These are the statistical reasons for each stage; the empirical sections test whether the combination delivers.

Stage 1 (parametric scale). Fit GARCH-t,

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by quasi-maximum likelihood on a trailing window, and keep only $\widehat{\sigma}_t$ and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Pickands, 1975; McNeil and Frey, 2000); the ES formula below requires $\xi < 1$, a condition every fitted tail in this paper satisfies by a wide margin and the code guards explicitly.[1] On losses $X = -\widehat{z}$, fix a threshold u at the empirical $(1 - p_0)$ loss quantile with $p_0 = 0.025$, fit the generalized Pareto distribution (GPD) $G_{\xi,\beta}(x) = 1-(1+\xi x/\beta)^{-1/\xi}$: shape ξ sets how heavy the tail is, scale β how wide) to exceedances $X - u > 0$ strictly on past data, and set, for $\tau \le p_0$,

$$\widehat{q}^{\,\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\,\mathrm{EVT}}(p_0) = -u$ (and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit; requires $\beta > 0$ and $1+\xi(x-u)/\beta > 0$). Since u is the *unconditional* empirical loss threshold while the Stage-2 body quantile at p_0 is the state-conditional $\widetilde{q}(p_0 \mid s)$, the splice is exactly continuous only when the threshold is anchored at the body's own p_0-quantile, $u = -\widetilde{q}(p_0 \mid s)$; otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join and the two pieces are spliced to agree in practice by construction. The tail is applied selectively where the raw tail under-covers (it hurts some volatility-index series and is not applied universally).

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$ and shift the curve per level by the empirical correction c_τ chosen as the $\lceil (n+1)\tau \rceil$-th order statistic of $\{e_i\}$: $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Romano et al., 2019). This distribution-free layer repairs the 97.5% level that every model in the battery otherwise fails.

Proposition 1 (split-conformal validity under exchangeability)**.** *If the calibration and test residuals $e_1, \ldots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \le n$ the shifted quantile satisfies*

$$\tau \;\le\; \Pr\big(z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\big) \;=\; \frac{\lceil (n+1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \ldots, e_{n+1}\}$ is uniform on $\{1, \ldots, n+1\}$; the event $z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \le e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \le \lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument, the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1), and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$, are in Appendix A. The guarantee is marginal, not conditional on s_{n+1}. □

The proposition's scope is narrower than its use. In deployment the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable; the proposition therefore motivates the construction, while the evidence that it works here is empirical: the exception-test repairs of Section 5 (the conformal stage is the only reason any entrant passes

[1] We use the peaks-over-threshold formulation (Balkema and de Haan, 1974; Pickands, 1975): exceedances over a high threshold converge to the generalized Pareto distribution, which uses tail data more efficiently than block maxima and is standard in risk management (McNeil and Frey, 2000; Embrechts et al., 1997); $\xi > 0$ is the heavy-tailed case.

Kupiec at 97.5%). The guarantee also attaches to the shifted quantile alone, not to the deployed curve of (5), whose tail branch takes the more conservative of the shifted and EVT values and is then monotone-rearranged; the exact finite-sample coverage identity need not survive that composition, so the coverage claim at the reported levels rests on these exception tests rather than on Proposition 1. Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018); we prefer the simple split form plus an empirical audit. One further implementation gap: the first-stage GARCH is fit on the full pre-test history, calibration window included, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer contests never read the calibration split and share their filter with the benchmark (identical information sets); the overlay's shift does use it, so the audit was rerun end-to-end strict, filter estimation stopped at the calibration boundary. Against the matched-information benchmark the advantage is intact (DM 5.1 and 5.4 at the two levels); the strict estimator's gap to the full-window benchmark (-0.6, -2.5) matches that benchmark's own margin over its short-window twin (2.8, 3.8), a quarter less estimation data, not leakage (replication artifacts: `job_fz_strict_calibration.py` $\rightarrow$ `fz_strict_calibration_results.json`). The static shift widens (per-name pass 34% against 48%), consistent with its disclosed conservatism. Run under this same strict split, the adaptive update (Gibbs and Candès, 2021) walks the shift from -0.51 to -0.20, lifts the per-name pass rate to 57%, and shrinks the joint-score gap to GARCH-t from DM -3.2 (static) to -1.5, not significant at the two-sided 5% level: the adaptive form is established as the remedy under the same information architecture. The estimator's annual-refit schedule sits between the static and adaptive poles, and the exception tests decide between them.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) \;=\; \widehat{\sigma}_t \cdot \mathrm{R}\Big[\; \min\{\widetilde{q}_\phi(\tau \mid s_t),\, \widehat{q}^{\,\mathrm{EVT}}(\tau)\}\, \mathbb{I}\{\tau \le p_0\} \;+\; \widetilde{q}_\phi(\tau \mid s_t)\, \mathbb{I}\{\tau > p_0\}\Big], \tag{5}$$

where $\mathrm{R}[\cdot]$ is the monotone rearrangement across levels. The tail forecast is the pointwise more conservative branch (the body branch is the minimum on 38% of tail nodes), and both risk measures come from this one curve. Recomputing ES_α as the numerical integral of (5) rather than the GPD closed form leaves every comparison in place (at 2.5%, FZ0 1.849 against 1.851, the integral marginally better, DM 5.3; estimator-over-GARCH DM 4.8 at both levels), and the splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$: DM 4.6–4.8).

Industry models as special cases. Equation (5) nests the standard toolkit. Historical simulation: $\widehat{\sigma}_t \equiv 1$ and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS): q the unconditional empirical quantile of past *residuals*, i.e. Stage 2 with the state dependence deleted. GARCH-t: $q = t_\nu^{-1}$, a fixed parametric shape. The hybrid nests FHS: whatever FHS captures, (3) can represent, and it adds conditional shape. (CAViaR sits outside the nesting: it autoregresses on the quantile directly, with no scale/shape split, and is the benchmark with which the estimator ties, Section 5.)

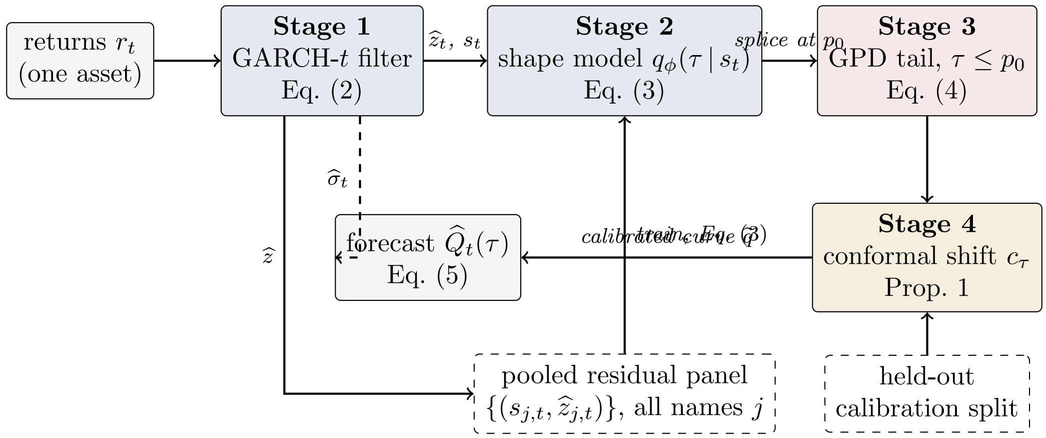

Figure 1: The residual-hybrid estimator as a pipeline. Solid arrows carry data; the dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve; dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal overlay). In production every stage refits annually using only past data; the evaluated protocol freezes each stage once per split, and the annual-refit rerun is reported separately.

Algorithm 1 (residual-hybrid estimator). (i) Fit (2) per asset on a trailing window; store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names; train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances; splice at p_0. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5). In *production* all stages refit on an annual walk-forward schedule; the paper's *evaluated* protocol freezes every stage once per split (the more conservative test, since no stage ever adapts to the test era), and Section 3.1 verifies the frontier separately under the true annual-refit schedule.

2.3 Estimating the shape: gradient boosting and neural quantile generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3); we use trees for point accuracy and the network when draws are needed (Table 1).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise, each new tree h_m fit to the negative gradient of the pinball loss (τ_k where the current model under-predicts, $\tau_k - 1$ where it over-predicts); the fitted grid is made non-crossing by monotone rearrangement, sorting the values $\{q(\tau_1|s), \dots, q(\tau_K|s)\}$ in τ, which never increases the loss (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest point estimator, about 0.75% better pinball than the tuned network, consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN; the GBC estimator). A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s, \tau) \;=\; g\big(\psi(s) \circ \varphi(\tau)\big), \qquad \varphi_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=0}^{n_c-1} \cos(\pi i \tau)\, w_{ij} + b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Its modifications, each forced by a documented failure, make it competitive on financial data: tail-aware τ-sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, oversampling the extremes that uniform sampling starves (the raw net's 99% breach rate was 3.2%); monotone rearrangement of the τ-curve; and the conformal layer of Section 2.2. The network's earning is not point accuracy but generation: with $\tau \sim U(0,1)$, $z^* = H_\phi(s, \tau)$ draws from the fitted conditional law by inverse transform. The network learns one continuous state-conditional quantile map, so draws are available directly, without separately fitting and interpolating a grid of quantile models (a GARCH-t or FHS model also simulates, from its fixed innovation law; the trees need the interpolation step).

How training works. The network minimizes the empirical version of (3) by stochastic gradient descent, each example drawn at a fresh level $\tau_{jt} \sim \lambda$ every epoch, so one set of weights carries the entire quantile curve. Early stopping is on held-out pinball loss; monotone rearrangement and the conformal shift are applied *after* training; and the whole pipeline is refit on the annual walk-forward schedule using only past data. Layer sizes, learning rates, and seeds are reported with the code release.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names; in generative-Bayes terms, H is an amortized map evaluated at any new (s, τ) without per-asset refitting. The intuition for why this works, and why it works at cold-start in particular, is actuarial. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns one universal mapping from observable state to residual shape ("assets whose trailing volatility looks like this, with characteristics like that, produce surprises shaped like so"), estimated from millions of asset-days of other names. A newly listed asset is not a new problem: its first ticks and characteristics place it at a point of state space already densely populated by other names' history, much as an insurer prices a new driver from a population table rather than from an empty claims file. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically. The same fact explains why the explicit own-history blends fail: the state vector already carries the name's own recent information, so bolting it on again adds only noise. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's own recent dynamics (realized vol above all; removing it costs +1.0%, removing all cross-sectional characteristics only +0.5%), with characteristics mattering at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update, both hurt at every listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 (benchmark tier) on the M5 competition data, outside finance altogether.

Practitioner's selection rule. The two estimators minimize the same loss over different index sets: the trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level; the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum, and the choice follows from what the application consumes. If the desk needs numbers at fixed regulatory

Table 1: Model selection by task and regime. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	§2.3
Single asset, calm (∼90% of asset-days)	GARCH-t	§4
Single asset, top misspecification decile	Nonparametric (+2.7%, DM 6.5)	§4
Amortized cross-name panel, any regime	Nonparametric always; no gate	§4.3
Regulatory ES with a ≥GARCH floor	Residual-hybrid + EVT + conformal	§5

levels (VaR/ES reporting, limits, capital), use the trees: best point accuracy, no GPU, no sampling needed. If the application consumes draws, scenario generation, portfolio aggregation by simulation, use the network: it learns one continuous conditional quantile map, so draws come by inverse transform with no grid interpolation, at a measured cost of ∼0.75% in point accuracy (a GARCH-t or FHS model also simulates, from its fixed innovation law; a tree grid requires monotone interpolation first). If both are needed, run both from the same pooled residual panel; they share every upstream and downstream stage of (5), so the swap is one component, not a new pipeline.

Algorithm 2 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t. (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z} - H_\phi(s,\tau))$ [network], or boost per-τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset, including one with no return history, using characteristics-only s, evaluate $H_\phi(s_t,\cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0,1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

2.4 Formal results: why shape misspecification must show up in the loss

The paper's organizing claim, that the nonparametric edge lives where the fixed innovation shape is locally wrong, is an empirical finding with a structural motivation in the pinball loss. The results below are deliberately modest: a shape violation creates a quantile wedge, and regret is locally quadratic in that wedge: motivation for looking where trailing shape diagnostics are extreme, not proof that mk_{63} estimates the wedge; that link is established empirically.

Lemma 1 (pinball regret identity)**.** *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ (so S is finite) and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(u) - \tau\big)\, du \;\geq\; 0, \tag{7}$$

with equality iff $F(u) = \tau$ for Lebesgue-almost every u between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof sketch. The difference quotients of $q \mapsto \rho_\tau(Z-q)$ are dominated by the Lipschitz constant $\max(\tau, 1-\tau)$, so differentiation under the expectation is justified and $S'(q) = F(q^-) - \tau$ (a.e. equal to $F(q) - \tau$); integrating from q_F to q gives (7) *for any* F, atoms forming a null set. The integrand $F(u) - \tau$ has the sign of $u - q_F$ (by definition of q_F), so the integral is ≥ 0 whichever side q lies; (8) then follows from $F(q_F) = \tau$ (supplied by the density hypothesis) and $m|u-q_F| \leq |F(u)-\tau| \leq M|u-q_F|$ for almost every u between q_F and q. Full details, including both orientations of the integral, are in Appendix A. □

Lemma 1 says the extra loss a misspecified forecaster pays is *locally quadratic in its quantile error*. The frontier claim then reduces to: does shape misspecification create a quantile wedge at the risk levels that matter? For the paper's canonical case, a fixed Student-t shape when the true local shape is fatter, it must:

Proposition 2 (tail wedge under tail-shape misspecification)**.** *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. There exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2)$ such that for all $\tau < \tau^*$ and $\nu_1 < \nu_2$, $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$: the fatter-tailed law has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}\,(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ at every level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof sketch. The unit-variance t_ν ($\nu > 2$, so the variance is finite and the normalization is well defined) has a regularly varying left tail: $\Pr(Z < -x) = c_\nu\, x^{-\nu}(1 + o(1))$ as $x \to \infty$ with $c_\nu > 0$; unit-variance rescaling alters c_ν but not the index ν. Inverting a regularly varying tail (Embrechts et al., 1997) gives $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$. For $\nu_1 < \nu_2$ the exponent $1/\nu_1 > 1/\nu_2$ makes $|Q_{\nu_1}(\tau)|/|Q_{\nu_2}(\tau)| \to \infty$, so the strict ordering and the diverging wedge hold for all τ below some $\delta > 0$; taking $\tau^* = \min(\delta, 1/2)$ places τ^* in $(0, 1/2)$ as claimed. (The proposition asserts existence of such a τ^*, not the location of the finite-sample crossover; the latter is $\tau_c \approx 0.0179$ for the t_3-vs-normal case of Remark 1.) The regret bound is (8); full details in Appendix A. □

Remark 1 (the crossover is real and matters)*.* Because unit-variance fat-tailed laws put more mass near the center as well as in the far tail, the quantile ordering of Proposition 2 reverses at moderate levels: numerically, the unit-variance t_3 has a less extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$ (solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root), so the deepest deployable level $\tau = 0.01$ sits below the crossover (fat tail strictly more extreme) while $\tau = 0.025$ sits just above it (at 2.5% the unit-variance t_3 quantile is -1.837, still less extreme than the normal's -1.960). Two of the paper's design choices are direct consequences. First, this is why CRPS-style whole-distribution scoring can rank a fat-tail-aware model below a thin-tailed one even when the tail is the decision-relevant region (Section 2.1): body and far tail disagree in sign. Second, it is why the deepest deployable level straddles the crossover rather than sitting cleanly beyond it, and why the score of Section 2.7 is used as a rank rather than a moment estimate.

The tail-wedge result of Proposition 2 motivates, but does not on its own establish, the empirical regularity that organizes the paper: that the nonparametric edge is ordered by the misspecification score. We do not claim a monotonicity theorem at the deployable levels, because two features of the standardized problem block one. First, Proposition 2 orders a given pair of tail indices only below a pair-specific level $\tau^*(\nu_1, \nu_2)$; it does not deliver monotonicity of the wedge in the tail index at a fixed operational τ. Second, and more fundamentally, the innovations are unit-variance by construction, and for unit-variance laws the level-τ quantile is not monotone in tail heaviness: as the tail fattens, the variance normalization pulls the fixed-τ quantile back toward the center (the crossover of Remark 1), so that at $\tau = 0.025$ a heavier-tailed t can carry the *less* extreme quantile. We therefore treat the score-ordering as an *empirical* regularity, documented below across the design panel, the frozen holdout, the FRTB battery, and the cross-market universes, and motivated by the tail-wedge mechanism rather than implied by it. The one exact statement we lean on is the regret identity (Lemma 1): wherever the benchmark's reported quantile differs from the oracle's at a level, it pays a pinball regret that is positive and, to second order, quadratic in that gap, and the score is our observable proxy for where that gap is largest.

Lemma 2 (validity of the generative sampler)**.** *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof sketch. For any x, left-continuity and monotonicity make $\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ an interval closed at its top, so $\{\widetilde{q}(\tau \mid s) \le x\} = \{\tau \le G(x)\}$ with $G(x) = \sup\{t : \widetilde{q}(t \mid s) \le x\}$. Since $\tau \sim U(0,1)$, $\Pr(Z^* \le x) = \Pr(\tau \le G(x)) = G(x)$, i.e. Z^* has CDF G. As G is the generalized inverse of $\widetilde{q}$ and $\widetilde{q}$ is nondecreasing and left-continuous, the generalized inverse of G returns $\widetilde{q}$ exactly (the standard inverse-of-inverse duality), so $\widetilde{q}(\cdot \mid s)$ is the quantile function of Z^*. Full statement in Appendix A. $\square$

Remark 2 (nesting and ERM dominance)*.* The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS; empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 5 test whether that dominance survives estimation noise. It does; the converse is that the same argument does not apply to CAViaR, which is why CAViaR is the one rival the estimator only ties.

2.5 A synthetic boundary check: when the frontier does not appear

Before the real data, a simulation with known truth (run once, seed fixed) checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the score: unit-variance Student-t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). First, the Lemma 1 identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 promises. Third, and most instructive, the score-binned nonparametric forecaster does not beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small; static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_ν law is a noisy statistic when tails are fat: its usual moment-based asymptotic variance requires a finite eighth moment ($\nu > 8$), and the population kurtosis itself does not exist for $\nu \leq 4$, so sixty-three observations cannot reliably rank regimes that differ only in a static ν. The jump and asymmetry signals, whose top deciles independently survive familywise control, corroborate the ordering, and a tail-index (Hill-type) variant of the score is an obvious robustness extension. For the empirical sections this means the real-data top-decile edge of Section 4 cannot be an artifact of stationary fat tails being averaged over (that configuration is the one the simulation shows a fixed shape handles) and is consistent with *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 negative control reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

2.6 Comparison with generative Bayesian computation

Relative to the generative Bayesian computation of Polson and Sokolov (2023), the downside-risk target changes four things in the estimator and one in the question. We train on a tail-weighted sampling measure $\lambda = \frac{1}{2}U(0,1) + \frac{1}{2}\mathrm{Beta}(0.3, 0.3)$ rather than $\tau \sim U(0,1)$, because uniform sampling starves the extreme levels downside risk depends on; we model GARCH-standardized residuals rather than raw returns, keeping the conditional-scale dynamics the parametric model already estimates well; we splice the GPD tail (4) at p_0 so that extrapolation beyond the training support is governed by extreme-value theory rather than the network; and the conformal stage adds an exchangeability-based coverage statement (Proposition 1). The amortization is also different in kind: we amortize across the real cross-section of hundreds of assets rather than across simulations of a single model, which is what produces the cold-start forecasts and transfer results of Section 2.3. Finally, the empirical question here, when such machinery improves on the parametric filter, is one generative computation does not address; the score below answers it.

2.7 The misspecification score

We use "misspecification score" as shorthand for a diagnostic of local residual-shape stress under the fitted GARCH-t benchmark; it is not a formal specification test.

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape. Static heavy tails do not violate this: ν absorbs them. What violates it is the shape being locally wrong: recent residuals far fatter-tailed or more asymmetric than any fixed t. We therefore score, per asset-day, over the trailing window $W_t = \{t-63, \ldots, t-1\}$ of $n_w = 63$ residuals; the window *ends the day before the forecast target*, so every signal is lagged at least one day relative to the loss it is paired with, and the frontier sorts of Section 4 pair S_{t-1} with the day-t loss (this alignment is enforced in the estimation code by construction) with window mean $\bar{z}$ and standard deviation s_z:

$$\mathrm{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \mathrm{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

In the released code mk_{63} and sk_{63} are the bias-corrected sample excess kurtosis and skewness (pandas' `kurt` and `skew`); at the full window length the correction is a common affine rescaling of (9), so the pooled decile ranks that the frontier uses are unchanged. Terminology is fixed here once: *skewness* is the signed third moment sk_{63}; *asymmetry* always means its absolute value $|\mathrm{sk}_{63}|$ (departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign); everywhere the text says "asymmetry," the statistic computed is $|\mathrm{sk}_{63}|$. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-5, \ldots, t-1\}} |\widehat{z}_i|$. The frontier score is the decile rank of these within the trailing cross-section. All three are predetermined (trailing-window) and computable on any asset with a fitted GARCH, which is what makes the frontier a deployable monitor rather than an ex-post classification. Informally we will call this the misspecification score; Table 1 reports the associated per-decile actions.

Why these statistics. Conditional on the fitted location-scale specification (μ_t, σ_t), the distributional restriction the flexible family relaxes is the fixed t_ν innovation *shape*. The lowest-order ways a fixed symmetric law can be locally wrong are precisely the third and fourth standardized moments: sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies (an exact t_ν sample has known positive excess kurtosis; the score is used as a rank, so only its cross-sectional and temporal variation matters, not its t_ν baseline). J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent, and Section 4 shows the realized edge concentrates there. Note the logical status of this choice: a signal that tracks departures from that fixed shape is a natural axis along which flexible-shape models would be expected to win, which the frontier evidence below and the t_ν-relative normalization together support.

Table 2: Sample flow across evaluation panels.

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map; engine development
Untouched-era holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	≈155,000	per-asset backtesting battery
Cross-asset sweep	Bloomberg	through 2026	43	[a]	frontier breadth
Country indices	Bloomberg	through 2026	26	[a]	frontier breadth

[a] Frozen exhibits; per-instrument windows vary and are documented instrument-by-instrument in the repository.

Algorithm 4 (the misspecification meter, as deployed). (i) Fit (2) on a trailing window; store $\widehat{z}_{t-62}, \ldots, \widehat{z}_t$. (ii) Compute mk_{63}, $|\mathrm{sk}_{63}|$, J_5 by (9). (iii) Convert each to its percentile against the trailing pooled panel (same asset class); the score is the maximum of the three percentiles, re-ranked into deciles. (iv) Act by Table 1: top decile $\Rightarrow$ the parametric model's average pinball loss runs ∼2–3% higher within this score region (a conditional average over the region, not a per-name point estimate), flag the asset for the nonparametric / residual-hybrid estimator and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 4.3). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

3 Data and evaluation protocol

Equity panels come from CRSP daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations; every modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: common shares (share codes 10 and 11) with at least 3,000 observations in the window, ranked by average market capitalization, top 200. Returns are in percent and include CRSP delisting returns where present; the one delisting event in the design panel moves headline results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data. Terminal access ended in mid-2026, so those exhibits are frozen at the version reported here and all continuing work runs on WRDS alone.

Licensed rows never leave the machines they were pulled to. The public repository carries code, configuration, and derived statistics; the panels rebuild from the queries above for any licensed subscriber. Table 2 gives the sample flow.

Splits are chronological within each name. The first 60% of a name's history is the estimation window and the last 40% is test; the final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH-t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on pooled estimation-window residuals, the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule's survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the conventions note that opens Section 4: mean pinball loss across eleven quantile levels for distributional accuracy, the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019) for joint (VaR, ES) pairs, per-date Diebold–Mariano statistics with a Newey–West variance for loss comparisons, the Model Confidence Set for multi-model fields, and Kupiec, Christoffersen, and date-clustered breach tests at the two regulatory levels for exception behavior.

"Frozen-specification" has a specific meaning in this paper, and we deliberately avoid the stronger word *registered*: no third-party registry or timestamp exists for these runs, so the claim is diffability, not notarization. A run is frozen-specification when the full specification (signals, features, hyperparameters, split rule, and test statistics) is fixed and the predictions are written down before the data are pulled. The holdout pipeline is line-for-line the design-era pipeline, so both files are public in the replication repository, so a reader can diff the two and verify that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 4 ran under this rule. As a post-hoc check, the deployed composite also replicates on this holdout (top decile +1.12%, DM 2.51; expanding prior-only cutoff +0.71%, DM 2.08), marginally above its kurtosis component (+1.04%, DM 2.16); because the composite was not the rule frozen at the time, this is replication on an already-examined sample, not a frozen-specification test. Frozen-specification runs that remain open are listed in Section 7.

3.1 Common shocks, calendar overlap, and the pooled learner

The design panels are split per name, 60/40 in each name's own time index. In an unbalanced panel such a rule can place one name's training window on another name's test dates, so a pooled learner could meet a systemic event in training that a per-name model would only meet out of sample, a leak that needs no individual feature to look ahead. We concede the point in principle: a per-asset split cannot by itself support a causal reading of a pooled edge. Here the channel is nearly degenerate (near-complete histories put split points on almost the same date), but nearly is not zero, and the question deserves an answer by construction.

The construction is a strict calendar-time split: every stage (each GARCH fit, the pooled learner, the GPD tail) is re-estimated on data strictly before 2020-01-01 and tested from that date for every name simultaneously, so no training window shares a single calendar day with any test window. The frontier survives: top decile +2.47%, per-date DM 6.22 over 1,258 dates (per-name split: +2.7%, DM 6.5); overall +0.71% (DM 1.87). The bottom decile is elevated here (+1.3%; a 2020–22 test era leaves few calm cells) but the ordering is intact: the top decile roughly doubles every other cell. The concentration is not a calendar-overlap artifact. A second cut targets the harder era: inside the 2000–2013 holdout, everything is re-estimated strictly before 2007-01-01 and tested on 2007–2013, so the global financial crisis is entirely absent from training for every name. The pooled learner still wins overall (+0.29%, per-date DM 3.08 over 1,762 dates) and the within-sample decile profile keeps its shape (top three deciles +0.45–+1.04% against +0.06–+0.09% in the bottom three); the frozen-threshold top cell of Section 4 is too thin in that era to test alone (3.4% occupancy, +0.85%, DM 0.7). The holdout spans the 2008 crisis, which lay in no training window, so the significant edge there is not an artifact of a calm era.

A last construction tests the schedule the paper recommends: the evaluated protocol freezes every stage once per split, so the score could in principle measure parameter staleness rather than shape. Under the true annual walk-forward (per-name GARCH refit on expanding data at each January-1 cutoff 2020–2024, residuals and score rebuilt under the refit parameters, pooled learner retrained) the top decile carries +2.47% at per-date DM 6.05 (lags 5–44: 6.6–5.4), indistinguishable from the frozen-fit frontier, while the overall edge is not statistically detectable (+0.32%, DM 0.8). Under the production annual refit the aggregate accuracy gap closes while the conditional high-misspecification advantage does not: refitting removes the average gap but not the concentration, so stale benchmark parameters are not what produce the conditional result.

3.2 Cross-sectional dependence, family-wise error, and the universe rule

Dependence, multiplicity, and survivorship receive the same treatment. *Dependence:* every headline test collapses the cross-section to one number per date before any variance is estimated (Section 4), so common-date dependence enters through the date series; bootstrap procedures resample dates, not asset-days. *Multiplicity:* the frontier grid spans three signals and ten deciles, and per-cell p-values ignore the thirty hypotheses' joint draw. We therefore report Romano–Wolf step-down family-wise-error control over the full 3×10 grid: per-date mean edges per cell, a stationary bootstrap over dates (mean block 10 days, $B = 2{,}000$; block 20 as sensitivity) that preserves cross-hypothesis dependence, and one-sided step-down adjusted p-values. Nine of the thirty cells survive 5% family-wise control under both block lengths, and they are the frontier's own: the top deciles of all three signals (mk_{63} $t = 9.4$, skew_{63} $t = 9.2$, jump_5 $t = 5.1$, adjusted $p < 0.001$), the ninth deciles of mk_{63} (adjusted $p = 0.022$) and skew_{63}, skew_{63}'s seventh, and jump_5's U-shaped far cells. Every mid-grid cell with an unadjusted t near two (mk_{63}'s sixth decile at $t = 2.34$, jump_5's seventh at 2.08, skew_{63}'s first at 2.01) loses significance under familywise control, and the headline claims rest on the cells that survive it. Any cell that loses significance under the adjustment is reported as statistically indistinguishable from zero in the text wherever it is discussed. The paper's guard against selection across its many exercises is replication, not one joint correction over everything reported: the headline is family-wise controlled within its grid, specified before the run and replicated on the untouched 2000–2013 holdout, and recovered under a strictly non-anticipating decile cutoff, so it does not hang on a single surviving test. The exploratory single-instrument universes, which are not jointly controlled with the equity grid, are labeled as such and carry their own 93-comparison adjustment. *Survivorship:* the holdout's observation filter conditions on names present for most of the era; the frontier comparison is relative (both models score the same names), so the filter tilts the universe toward survivors without by itself favoring either model in the ranking. The sharper version of the objection is selection on information unavailable at the start, and it too has a constructive answer: a point-in-time re-selection fixes the universe at the top 200 names by market capitalization as of the first quarter of 2000, applies no minimum-history filter, and keeps every name through its delisting date with the CRSP delisting return appended as the final observation. The point-in-time construction addresses survivorship directly: on the 164 of 200 Q1-2000 names that clear the symmetric fitting floors (36 dropped; 79 delisting returns ridden to the end), the overall edge is +0.58% with per-date DM 8.96 over 2,692 dates, and the frozen design-era top bucket carries +2.94% at DM 6.6 (5.9% occupancy; the bottom bucket is +0.44%, DM 5.4, in this era even calm cells lean positive, at one-seventh the top bucket's size). The edge is larger than in the survivor-tilted holdout, so the observation filter biased the reported edge downward rather than upward. This point-in-time construction still requires the symmetric fitting floors, so 36 of the 200 names, those the estimators cannot fit, do not enter; the estimand is the 164 that clear them.

3.3 Reporting and inference conventions

The reporting and inference conventions used in the empirical sections that follow are fixed here. (i) *Edges* are percentage reductions in pinball loss, so *larger is better* for the named model; a "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. (ii) *Ratios* are model-over-benchmark loss ratios, so *below 1 favors the model* and $1 -$ ratio is the edge (0.973 = a 2.7% win; 1.043 = a 4.3% loss). (iii) *DM* is the Diebold–Mariano t-statistic on the loss difference. *Per-date DM* means, concretely: losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$ with the orientation stated at each use (benchmark minus engine in the engine's headline comparisons, so positive favors the engine), and the test is run on the resulting $\sim$1,100-date series with a Newey–West (lag-10) variance: the effective sample is the number of dates, never the number of asset-days. Reported p-values are *one-sided* in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%); t-statistics above 2.6 are significant under either convention, and marginal cases are flagged where they occur. A tie means the per-date DM cannot reject zero at the two-sided 10% level; the point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible we make it as one: for the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin, a two-one-sided-tests conclusion, not a failure-to-reject. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic and a stationary bootstrap (mean block length 10 days; $B = 1{,}000$ replications in the main battery, $B = 800$ in the CAViaR extension), following Hansen et al. (2011). (iv) *Coverage* numbers are realized frequencies against a stated nominal (0.05 target:

Table 3: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing misspecification score (200 CRSP names, 221.6k test asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\lvert\text{sk}_{63}\rvert$ (asymmetry)	jump_5
1–8 (bulk)	≈ 0	≈ 0	mixed, ≈ 0
9	+0.52	+0.69	—
10 (top)	+**2.46**	+**2.37**	+0.90
Overall		+0.44	

Per-date Diebold–Mariano: top mk_{63} decile edge +2.71%, DM 6.54 ($p \approx 0$); bottom decile +0.19%, DM 1.41 ($p = 0.079$, not significant); top two deciles pooled +1.65%, DM 6.74; overall +0.44%, DM 5.67. Top-decile signal means: kurtosis ≈ 25, $\lvert\text{skew}\rvert \approx 4$. Three top-decile conventions appear in this section and differ by construction: pooled asset-day weighting gives +2.46 (the table cells), equal-date weighting gives +2.71 (the DM convention), and the timing-diagnostics run of Figure 3 reports +2.69 under its deployed one-day lag.

0.056 is slight under-coverage of risk, 0.032 over-conservatism); for *credible-interval* coverage, closer to nominal is better in both directions. (v) *Correlations* quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

4 The misspecification frontier

4.1 The within-universe result

On 200 CRSP names ($\approx$221.6k test asset-days), we bucket the per-day pinball edge of the amortized nonparametric model over GARCH-t into deciles of each misspecification signal. Table 3 shows the result that organizes the paper.

The edge is a top-decile phenomenon. It is large and significant where the misspecification score is high, and statistically indistinguishable from zero where it is not.

The deployed composite. The monitor the paper deploys is not mk_{63} alone but the composite score, the maximum across the three signals' trailing-panel percentiles (kurtosis, asymmetry, jump). Evaluated directly, its top decile reproduces the frontier at +2.79% (DM 8.9), a clean 10% cell whose lower nine deciles sit between −0.3% and +0.35%, and it survives the strictly non-anticipating expanding cutoff at +2.72% (DM 9.0). The three components each carry a top-decile edge on their own, +2.98%, +2.78%, and +1.96% (kurtosis, asymmetry, jumps); the maximum of decile ranks instead flags the union of the three top deciles, a 19% cell at +1.72%, so we report the re-deciled composite.

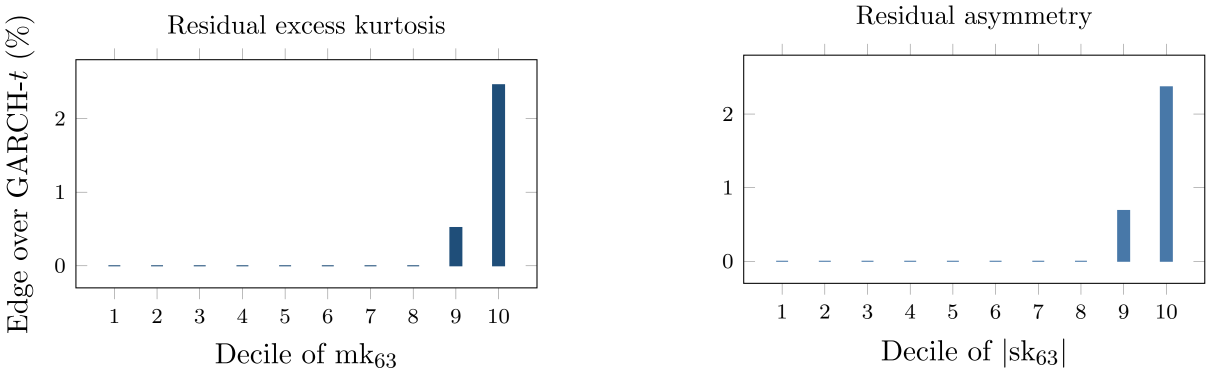

Figure 2: The misspecification frontier. Nonparametric pinball edge over GARCH-t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

Does the scoreś content come from model-relative misspecification, departure from each nameś own fitted t_{ν_i}, or simply from the absolute level of recent tail activity? Converting each mk_{63} into a percentile of the simulated null of 63-window kurtosis under the name's fitted t_{ν_i} leaves the frontier essentially unchanged (top decile +3.0%, DM 8.5, against +3.0%, DM 10.5 raw), but this normalization is nearly rank-preserving (Spearman 0.97, 82% top-decile overlap) and so cannot by itself separate the two. A discriminating test instead favors the raw-tail-activity reading. In a within-date cross-sectional (Fama–MacBeth) regression of the per-observation edge on the raw score and the ν-relative percentile, the raw score carries the predictive content ($t = 7.8$) while the ν-relative percentile has significantly negative incremental content ($t = -5.2$); double-sorting agrees, the top-raw-kurtosis-quintile edge falls from +1.9% for the most fat-tailed (low-ν) names to +1.5% for the thinnest-tailed, and observations high in the ν-relative score but not in raw kurtosis carry no edge (+0.3%, DM 0.5). The signal is thus the absolute level of recent tail activity rather than a per-name ν-normalized surprise. The signal still measures shape stress relative to the fitted model: mk_{63} is computed on the standardized residuals the model treats as stationary t_ν, so elevated values flag states in which the realized residual shape is heavier or more asymmetric than the fitted law typically produces. A correctly specified low-ν t_ν can itself generate high realized kurtosis, so the score is a stress indicator rather than a formal specification test; what the data reject is the finer per-name normalization, plausibly because a name's fitted ν is estimated from the same residuals and a low value is itself a symptom of the tail activity that matters, so dividing it out removes part of the signal. The flexible estimator's advantage concentrates wherever recent standardized-residual tail activity is high.

Timing diagnostics. The score is computed from residuals through day $t-1$ and paired with the day-t loss; Figure 3 probes that alignment. The deployed lag gives the largest top-decile edge (+2.69%, DM 6.5); a five-day-old score retains +1.67% (DM 6.4). A deliberately contaminated

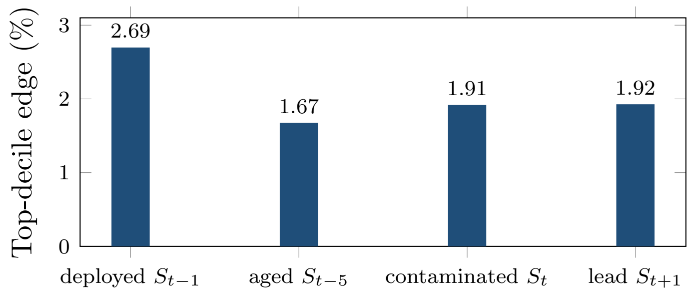

Figure 3: Timing diagnostics for the frontier. Top-decile edge under four alignments of the mk_{63} score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less, the opposite of what a look-ahead artifact would produce.

same-day window, the alignment a look-ahead artifact would favor, yields less than the deployed lag (+1.91%; a one-day-ahead placebo gives +1.92%). Same-day sorting does not manufacture the edge.

The decile boundary is non-anticipating too. The sort forms deciles once over the pooled test panel, so top-decile membership draws on the full-sample distribution of mk_{63} even though the score itself is non-anticipating. Re-forming the 90th-percentile cutoff at each date from only mk_{63} observed on strictly earlier dates (250-day burn-in, the rule the deployed monitor uses) leaves 86% of membership unchanged and holds the top-bucket edge at +2.45% (DM 8.3), essentially the headline: the pooled-panel cutoff is not what produces the concentration. A within-date cross-sectional decile, which is not the deployed rule and forces 10% occupancy on every date including calm cross-sections with no misspecified name, dilutes the edge to +0.26% (DM 2.2). The frontier ranks the absolute level of residual misspecification, not within-day position.

An absorption test. If the score were merely an omitted variable, a parametric-adjacent fix should capture the edge without a pooled learner: per-name FHS with the residual window reweighted toward days whose mk_{63} resembles today's (Gaussian kernel). On the full panel the reweighted FHS is significantly worse than plain FHS overall (per-date DM -11.3) and in the top decile (pinball 0.541 against 0.444), while the pooled model beats plain FHS there by 2.7% (DM 8.8). The deterioration is consistent with effective-sample collapse: sharp univariate conditioning shrinks a name's usable history exactly when today is unusual. Pooling across names is what makes state-conditioning affordable: the amortized model borrows the shape information from every name's stressed days. These results suggest the amortization of Section 2.3 is important for the frontier result: a single name's history cannot support sharp state-conditioning on its own.

Table 4: Double sort: the edge needs both a shape break and an active regime. Cells are the nonparametric-over-GARCH-t pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

	rv_{63} Q1 (calm)	Q2	Q3	Q4	Q5 (active)
mk_{63} Q1 (low)	+0.24	+0.34	+0.10	+0.01	−0.14
Q2	+0.16	+0.20	+0.22	+0.09	−0.57
Q3	−0.02	+0.19	+0.25	+0.44	+0.04
Q4	+0.08	+0.15	+0.35	+0.23	+0.10
Q5 (high)	−0.10	+0.44	+0.68	+0.61	**+2.33**

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4. The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

Robustness of the score. Table 4 double-sorts the edge on the score and on trailing volatility. Sorting test asset-days independently into quintiles of mk_{63} and of trailing volatility (rv_{63}): high volatility without a shape break carries no edge (top-volatility column, bottom four kurtosis quintiles: −0.57% to +0.10%), a shape break without an active regime carries almost none (top-kurtosis row, calmest quintile: −0.10%, DM 0.7), and the product state carries +2.33% with DM 6.4. The two ingredients interact. The score flags a currently wrong parametric shape; volatility scales the cost of the error. Second, a superior-predictive-ability test (Hansen, 2005) on the per-date loss panel against the full desk field (GARCH-t, FHS, EWMA, HS) returns $T^{\text{SPA}} = 0$, consistent $p = 1.00$: no rival's advantage is even positive, and each is individually worse with $t \geq 5.8$. Third, the score needs no recalibration to stay useful: holding the design-era top-decile threshold fixed, the top-decile edge is positive in every test year (between +1.7% and +3.4%, 2020–2024), even though the cross-sectional 90th percentile of mk_{63} itself drifts upward in 2023–24. Fourth, the score is a slow state, so monitoring it costs little: a median name enters the top decile less than once a year (0.68 entries) and a median episode lasts 24 trading days.

The untouched-era holdout. Every design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze the entire pipeline (signals, features, hyperparameters, split rule, and test statistics), wrote down three predictions in advance (top-decile edge positive and significant; bottom decile indistinguishable from zero; a threshold-shaped decile profile), and then pulled CRSP daily returns for 2000–2013, an era no prior run in this project had touched. The frontier replicates (Figure 4): the top-decile edge is +0.89% with per-date DM 2.16 ($p = 0.016$; 1,465 dates), deciles one through nine sit between −0.18% and +0.28%, and the overall edge is +0.02%: the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. The holdout spans the 2008 crisis, so the replication is not an artifact of a calm era. One measurement choice in the replication needs a check: decile boundaries above are within-sample ranks of the holdout test era (the standard conditional-sort convention), so the partition rule, though it never touches a forecast or a loss, is not a rule a desk could have applied on day one. The day-one-applicable restatement freezes the design era's numeric decile

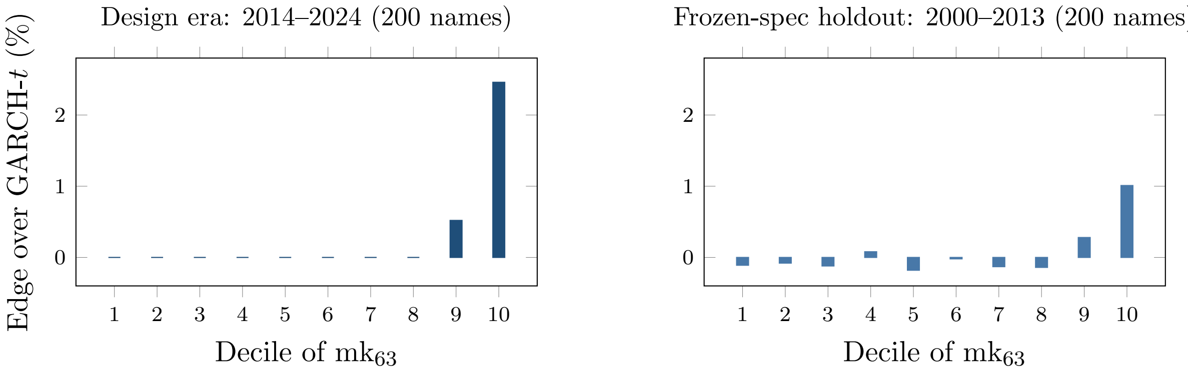

Figure 4: The frontier out of era. Left: the design-era decile profile (Table 3). Right: the same pipeline, frozen and run once on 2000–2013 CRSP data under the frozen specification. Bars are pooled per-observation decile means (top decile +1.01%); the inference statistic is the per-date aggregation of Section 2.7, +0.89% with DM 2.16 over 1,465 dates. All other deciles sit between −0.18% and +0.28%. The threshold shape survives an untouched era containing the 2008 crisis.

boundaries (the pooled design test sample's mk_{63} edges, top threshold 13.67) and apply them to the holdout as constants, so membership at (i, t) depends only on name i's own past 63 days and a number fixed before any holdout day was scored. The threshold shape survives the frozen rule (bins eight through ten carry +0.82%, +0.92%, +1.02% against −0.3–0.0% below), and the top-bin point edge is slightly larger than the within-sample sort's (+1.02% vs +0.89%), but the cell thins to 3.3% occupancy in this lower-kurtosis era and the per-date test loses power (DM 1.11, insensitive to Newey–West lags 5–44). A second real-time rule needs no design-era constant at all: an expanding pooled 90th-percentile threshold over strictly earlier dates (250-date burn-in) restores natural occupancy (9.5%) and recovers the sort's result with significance: +0.99%, DM 2.15, insensitive to Newey–West lags 5–44; the bottom bucket is −0.18% ($t = -1.4$). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge; the recursive rule is the version a desk could have run from day one (the frozen thresholds' own decisive test is the point-in-time universe of Section 3.2: +2.94%, DM 6.6).

The overall signal–edge correlation is low (0.05–0.10) because the relationship is a threshold nonlinearity, not a linear one; the top-decile Diebold–Mariano statistics are the appropriate summary.

4.2 The frontier across universes

Table 5 shows the same axis at work across markets. First, the only decisive standalone single-asset wins anywhere are the hyperinflation and capital-control currencies, Argentine and Egyptian pesos, 12–14% better with significance, whose residual kurtosis (∼2000–3300) sits two orders of magnitude above anything GARCH-t was designed to absorb. Second, the country gradient is mediated by residual kurtosis, not by the development label per se: frontier indices (kurt ∼10) sit between well-behaved developed indices (kurt ∼5) and hyperinflation FX on the same frontier. Third, the Korea-2026 negative control sharpens the mechanism: a dramatic price

Table 5: The frontier across universes. Ratio = nonparam/GARCH-t pinball (< 1: nonparametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM -13.7
	EM (10)	1.013	8	GARCH, pooled DM -9.5
	crisis (7)	0.973	$\sim$1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86, $p = .002$
	USDEGP	0.857	1920	DM 3.80, $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins[a]
Cross-asset (43)	vol indices	$\sim$0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
13/43 significant wins overall; corr(log kurt, edge) = 0.43				
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM -2.5 to -5

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06); exploratory: only Kenya survives the 93-comparison familywise adjustment. Cross-index corr(log residual kurtosis, edge) = 0.53. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating that kurtosis is neither necessary nor sufficient.

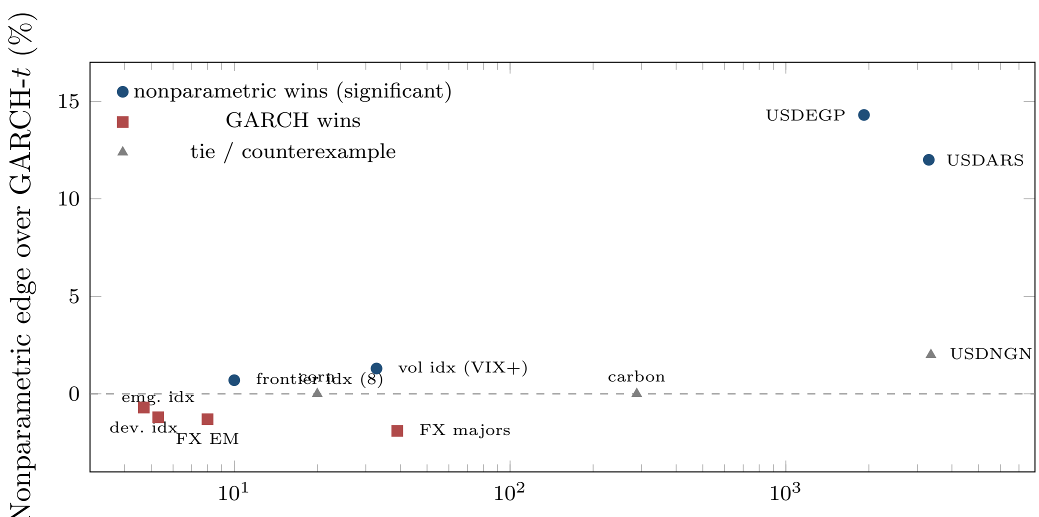

Figure 5: The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis $\sim$2000–3300) carries the only decisive standalone wins; high kurtosis is predictive but neither necessary nor sufficient: carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

crash with circuit breakers is not residual misspecification: Korean residual kurtosis stays in the ordinary 4–13 range, and GARCH-t significantly wins all twenty-three assets even inside the crisis window. The frontier is about the residual shape, not about headline turbulence.

One adjustment prices the breadth of this search. The three exploratory universes contain 93 instrument-level comparisons; treated as one family, a Holm step-down at familywise 5% keeps eight numerical survivors, of which one (PJM power) is already reclassified by the audit below as a log-return artifact on near-zero prices, leaving seven: the three volatility indices, Baltic freight, wheat, USDEGP, and Kenya. Sri Lanka, Argentina, Pakistan, and Nigeria drop from the individually significant list. Instrument-level entries are therefore exploratory; the replication evidence this section rests on is pooled and structural: the crisis-tier FX comparison (DM 4.12, $p \approx 2 \times 10^{-5}$, surviving any adjustment) and the kurtosis–edge gradients across universes.

A frequency corollary. In the five daily crypto series examined here, GARCH wins throughout (zero nonparametric wins). At *hourly* frequency the picture inverts: the neural IQN beats the full benchmark ladder, including filtered historical simulation, by $\sim$0.6% with per-date DM $p < 10^{-7}$. Sample density is itself a frontier dimension: more observations per unit of regime raise the resolvable residual-shape signal. A systematic intraday map of the frontier is pre-committed and blocked only on tick-data access.

4.3 Regime dependence and model selection

In the amortized panel the edge concentrates in turbulence: the GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent (roughly fourfold) while both nonparametric estimators beat GARCH in every quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478). A natural inference is that one should gate: run GARCH on calm days and switch to the nonparametric model when the score is high. Gating does not improve on continuous use of the flexible model. A classifier gate trained on the residual-shape signals routes 90.5% of days to the nonparametric model and exactly matches always-nonparametric (0.3987 vs GARCH 0.4000); worse, its routing runs backwards across score deciles, because nonparametric wins on high-misspecification days are rare-but-large while a classifier targets frequency. A regression gate on the edge magnitude fixes the direction (routing flat ≈ 1.0, predicted-edge correlation +0.16) and still converges to always-nonparametric. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by which day the tail event lands, and that is unpredictable from prior-day state. The deployable conclusions: in the amortized setting, use the nonparametric model always (it does not underperform on calm days); use the residual-hybrid where a guaranteed $\geq$GARCH floor is wanted; and use the frontier score as a monitor (Section 6); day-level switching adds nothing. (Detection latency is small, median zero, and the edge is back-loaded within episodes, so the lag costs little even in the per-name setting.)

Table 6: FRTB battery: average pinball loss (twelve quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	$Breach_{99}$	$Kupiec_{99}$ p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	−6.37 / −5.83	1.09%	0.00
+ EVT tail	0.3555	−6.92 / −5.91	0.91%	0.00
GARCH(1,1)-t	0.3561	−6.69 / −5.65	1.06%	0.026
GJR-GARCH-skew-t	0.3562	−6.58 / −5.84	1.05%	0.062
GARCH-FHS (per-name)	0.3562	−6.61 / −5.84	1.11%	0.00
GARCH-FHS (pooled)	0.3566	−6.90 / −5.67	0.99%	0.79
GARCH-FHS (rolling 500d)	0.3569	−6.53 / −5.98	1.09%	0.00
EWMA (RiskMetrics)	0.3606	−5.52 / −5.63	1.63%	0.00
Historical simulation	0.3619	−7.21 / −6.86	0.84%	0.00

All columns are generated from a single common-sample run: `frtb_table_results.json` on a single common sample (155k asset-days; the rolling-FHS burn-in trims all models identically). Predicted ES is the numerical tail integral $\alpha^{-1}\int_0^\alpha Q(u)\,du$: closed forms for the t and normal entrants, 200-node integration of the Hansen skew-t inverse CDF, and a matched 20-node midpoint rule for the empirical quantile models and for the hybrid engine, whose VaR and ES are both read from one monotonized min-envelope curve $Q^* = \min\{\text{body},\text{EVT}\}$. An earlier draft approximated predicted ES by a three-node tail-quantile average, understating it by 2–6%; the numerical integral is why these levels differ from prior versions. Diebold–Mariano vs the hybrid: GARCH-t 5.0, GJR-skew-t 5.0, FHS pooled 6.2, per-name 5.2, rolling 5.7, HS 7.5, EWMA 8.7 (all $p \approx 0$; rounded, and stable across reruns); the EVT variant concedes 0.0004 pinball to the raw hybrid (DM $\approx$6) as the price of its tail; the 90% MCS contains the raw hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a diagnostic, not a ranking: "realized" is the mean return below each model's own VaR, so the conditioning set is model-dependent, and under the numerical integral every fat-tailed entrant exceeds its own-tail comparator by 9–22%. The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew-t row: corrected Hansen inverse CDF with the fitted η, λ (median 4.2, −0.015; validated against the symmetric-t limit and `arch`'s own distribution), the original implementation silently reduced to a symmetric $t(8)$; defect and correction are disclosed in the availability section.

Exception columns are pooled panel diagnostics: at 155k observations a breach-rate deviation of ± 0.1 points rejects in either direction (the EVT variant's 0.91% sits on the conservative side; pooled FHS happens to land at 0.99%), so pooled p-values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement (Figure 7): per asset the EVT-tailed engine passes Kupiec at 99% for 84% of names and Christoffersen for 86%, and the date-clustered test passes at both levels ($t = 1.01$ at 99%, 1.86 at 97.5%) while the raw hybrid is rejected ($t = 4.5$): the tail stages carry the regulatory calibration, and the 2008-era battery re-run confirms it out of era. At 97.5% the split-conformal recalibration is the stage that repairs pooled Kupiec (GBM-recal $p = 0.24$).

5 VaR and Expected Shortfall forecast evaluation

5.1 Head-to-head accuracy and significance

Table 6 reports the full battery on 140 CRSP names (155k obs, 1108 dates).

The stress-era replication. The battery's pre-committed re-run on the 2000–2013 holdout panel is complete. Splits follow Section 3, so each name's test window runs from roughly mid-2008 through 2013, the crisis and its aftermath. The EVT-tailed engine passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t-statistics of 0.17 at 99% and

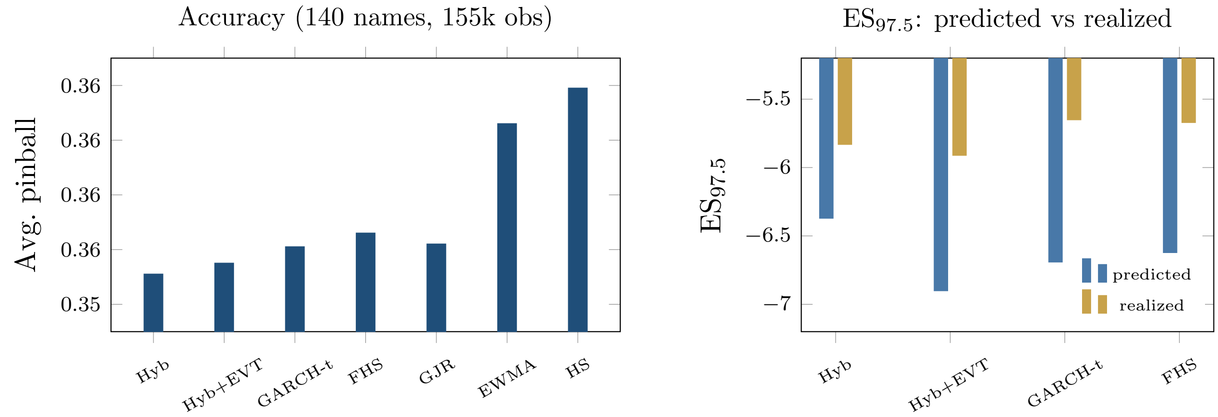

Figure 6: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 5.2–8.8 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted (exact tail integral) vs realized below each model's own VaR, a model-dependent diagnostic, not a ranking (every fat-tailed entrant's integral exceeds its own-tail mean); the joint FZ0 score is the ranking criterion.

0.35 at 97.5%, consistent with nominal at both regulatory levels through the crisis window. The two models most common on desks do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH-t is also well calibrated in this era (94% pass rate), which is the frontier's own prediction: calibration parity is the norm, and the engine's added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative out of era (breach 2.21% at 97.5%, $t = -1.09$, not significant): the safety margin widens when the calibration era is calmer than the test era, and the direction of the error is the one regulators prefer.

Adding SAV-CAViaR, a model absent from banks' standard toolkit but the strongest semi-parametric academic benchmark, produces a statistical tie: CAViaR 0.3461 vs hybrid 0.3460 (DM 0.4, $p = 0.34$; the CAViaR study's common sample differs slightly from Table 6's, which is why the hybrid's loss level is 0.3460 there against 0.3551 in the table; rankings, not levels, are comparable across the two runs); the 90% Model Confidence Set contains exactly these two. The hybrid is *co-best with CAViaR*, not uniquely dominant, and still significantly beats everything banks actually run. SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history (no new listings), where the hybrid produces the full curve and prices new listings. We do not hold pooled exception rejections against it: pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, ours included; the date-clustered restatement is the appropriate reading. The accuracy tie is between a single-level autoregression and one shared model carrying the whole deployment brief. Restating FHS the textbook way changes nothing: per-name train-constant and per-name rolling 500-day trailing variants, run on the identical panel, lose to the hybrid by DM 5.4 and 5.8 (pinball 0.3562 and 0.3569 against 0.3550), bracketing the pooled variant the

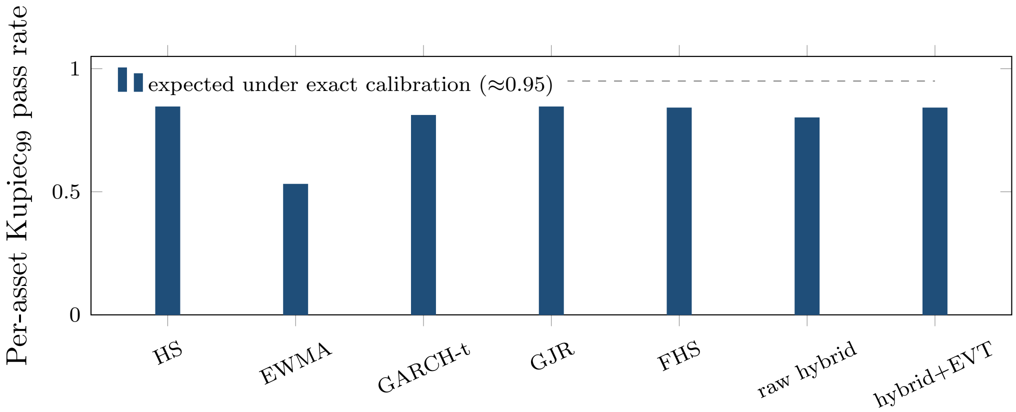

Figure 7: Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and the joint FZ0 score (Table 6). Date-clustered tests in the table notes.

battery uses. Second, the neural IQN as residual model initially loses to trees badly (0.3650 vs 0.3551, DM 13.3) with a severely thin tail (breach$_{99}$ 3.2%); the tail-aware, monotone, conformally recalibrated version closes to a hair (0.3638 vs 0.3632, DM 3.03) with calibration repaired (breach$_{99}$ 0.92%). Trees remain the strongest tabular estimator; the tuned IQN is competitive. Third, the ES columns are a diagnostic, not a ranking. Under the exact tail integral (a numerical correction: the earlier three-node tail average understated predicted ES by 2–6%), every fat-tailed entrant's predicted ES exceeds the realized mean below its own VaR, by 9% (raw hybrid, rolling FHS) to 22% (pooled FHS), and the comparator conditions on each model's own breach set. The joint FZ0 re-scoring below is the ES ranking this paper stands on.

The joint (VaR, ES) **score.** Because ES is not elicitable alone but the pair (VaR, ES) is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the battery was re-checked under the FZ0 joint score (orientation validated by Monte Carlo against a known-truth t_5 before any ranking was read; first sandbox subsets of the panel, full-panel re-run pre-committed). On a 45-name large-cap subset, a Gaussian-innovation GARCH is significantly worst at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, fat tails are strictly necessary. Among the fat-tailed entrants (GARCH-t, FHS, and the hybrid+EVT) FZ0 differences on large caps are statistically indistinguishable ($|\mathrm{DM}| < 1.4$): on the well-behaved segment the EVT stage's distinct value is its exception-test compliance, a calibration property the FZ0 score does not reward. On a 40-name top-kurtosis subset the ordering flips and the hybrid+EVT attains the lowest FZ0, the frontier pattern reappearing under a jointly consistent score, directionally but not yet significantly at this width (DM ≈ 0.9, $p \approx 0.17$).

On the full 200-name panel (221.6k test asset-days), with VaR and ES read from the coherent min-envelope curve Q^* and per-date DM inference throughout, the engine's *accuracy layer*, the EVT-tailed forecaster with no conformal shift, attains the lowest FZ0 at both regulatory levels: at 1% it beats GARCH-t (DM 4.8) and FHS (DM 4.9), the two standard benchmarks the accuracy claim rests on. A per-name score-driven one-factor GAS model estimated by FZ0 minimization in the style of Patton et al. (2019) loses by more (DM 9.6 at 1%, 6.2 at 2.5%), but we read that comparison as indicative only: a few names' GAS optimizers destabilize at 1% and inflate the gap, while a three-start re-estimate settles the stable 2.5% comparison near DM 5.9. At 2.5% the engine again wins over GARCH-t (DM 5.1); the accuracy layer is thus a single unified configuration: lowest FZ0 at both levels and a pass on the aggregate date-clustered exception test at both levels, with per-name diagnostics showing residual cross-sectional calibration heterogeneity (84%/86% pass rates at 99%; Figure 7), with only the hypersensitive pooled 155k-observation Kupiec at 97.5% against it. The frontier concentrates the joint-score edge as it does the pinball edge: the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.6% against the 2.5% target) and gives back the FZ0 advantage there: the static-overlay variant loses to GARCH-t on the 2.5% joint score (DM -1.6, not significant at the two-sided 5% level), and the concession is largest exactly where the bands are widest: in the top misspecification decile the static shift's joint-score deficit reaches DM -6.7. The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative (wider bands, fewer breaches than nominal). A desk that prizes the conservative coverage margin keeps the shift; one scored on FZ0 efficiency runs the EVT tail alone, which passes the exception battery on its own (Figure 7). The concession is mostly staleness, not structure: the adaptive form of the shift (Gibbs and Candès, 2021), updated daily from the panel breach frequency, walks the margin back out of era (-0.35 to -0.10; breach 1.8% $\rightarrow$ 2.1%) and erases the overall concession: the adaptive variant ties GARCH-t on 2.5% FZ0 (DM -0.3) where the static loses (DM -2.0, same run). The top-decile cost survives any shift (-4.8 vs -6.0): the irreducible price of a margin where bands are widest. The static shift also buys per-name coverage badly (Kupiec$_{97.5}$ pass rate 47% against 72% unshifted, 69% adaptive; over-coverage fails from the other side). Every "lowest FZ0" headline refers to the accuracy layer; the static concession is a frozen margin's price, and the adaptive form removes the overall cost.

5.2 The ten-day horizon: a direct multi-day extension

The multi-day results in this subsection come from a direct amortized model, a pooled quantile learner trained on h-day cumulative returns with GARCH-derived features, and not from the residual-hybrid estimator of Algorithm 1, whose residual architecture is one-day; the two are separate objects, and no ten-day claim in this paper attaches to the deployed engine. Training labels whose h-day window crosses the split are excluded, so no label contains test-era returns. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) the direct model's pinball edge over $\sqrt{h}$-scaled GARCH grows from $\sim$0.4% at $h = 1$ to $\sim$1.4% at $h = 10$, and holds in the 2020–2022 stress window (hybrid 1.248 $<$ FHS 1.2605 $<$ GARCH 1.262). The design's h-curve (Figure 8) rises from 0.45% at h=1 to 2.1% at h=20 under the same purge.

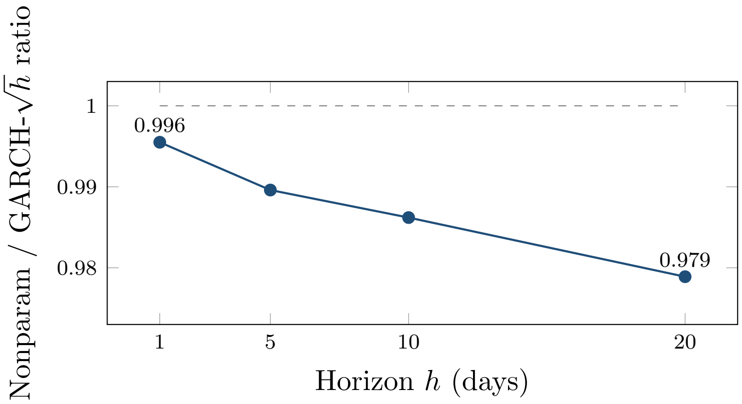

Figure 8: The edge grows with horizon. Direct-model/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (150-name design panel; training labels whose h-day window crosses the split are purged): fat tails compound while $\sqrt{h}$ scaling compounds the shape error, from 0.45% at h=1 to 2.1% at h=20. The forecaster is the direct multi-day model of the main text's ten-day subsection, not the residual-hybrid estimator, whose architecture is one-day. An earlier version of this figure showed larger ratios from an unpurged run; the purged rerun supersedes it.

More consequentially for tail severity, with ten-day ES now the same exact tail integral as Section 5, and the same model-dependent own-tail caveat attached: every ten-day tail over-states its own-tail comparator, GARCH $\sqrt{h}$ most (predicted −21.1 vs realized −15.4) and the direct model least (−20.1 vs −17.3), with the best ten-day breach$_{97.5}$ (2.79% vs GARCH 3.30% against the 2.5% target). Fat tails compound with horizon; $\sqrt{h}$ Gaussian-style scaling compounds the shape error. A caveat: the temporal split places the whole test window in 2020–2024, and the pre-committed rerun on 2000–2013 shows the ten-day result is era-dependent. Out of era the two methods tie on accuracy (direct edge +0.49%, date-level DM 0.07, NW lag 10), and the calibration verdict reverses: $\sqrt{h}$-scaled GARCH-t ES is almost exactly calibrated through the crisis window (predicted −16.6 vs realized −16.6, numerical integral, purged) while the direct model's tail under-states it (−15.9 vs −17.8): the pooled ten-day tail, trained on the pre-crisis segment, did not extrapolate to 2008-scale ten-day losses. The ten-day tail-severity advantage is therefore a 2014–2024 statement, and a desk holding the engine at $h = 10$ through a 2008-type era should keep the parametric $\sqrt{h}$ number alongside as the conservative envelope. The frontier itself is confirmed on 2000–2013 (Section 4); this horizon result completes the pre-committed comparison with a mixed verdict, reported as such.

5.3 Cross-asset evidence

Across the 43-instrument sweep the relationship is heterogeneous. The one calibrated equity-volatility gain, VIX, is marginal (one-sided $p = 0.041$) and does not survive the multiplicity adjustment; larger accuracy gains on credit and Baltic freight fail breach-independence and are reported as accuracy edges with open calibration rather than wins; and an apparent electricity win reversed once the return transformation was corrected for near-zero prices. The GPD tail also hurts some volatility indices, motivating the selective-application rule of Section 2.2.

Instrument-level results and diagnostics are collected in the Online Appendix.

6 Extensions and implications

The production loop. *Nightly:* (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau \mid s_t)$, one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and, where the desk elects the coverage overlay, the conformal shift c_τ (static or adaptive); (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\sigma}_{t+1}\,\widetilde{q}(\tau \mid s_t)$ at the desk's levels, with VaR and $\mathrm{ES}_{97.5}$ both read from the coherent min-envelope curve $Q^* = \min\{\text{body,EVT}\}$ (VaR at the α node, ES its numerical tail integral); (v) read the misspecification score (9) as the model-risk monitor: top decile means the parametric fallback's conditional average loss in that score region runs ∼2–3% higher. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* run the separate characteristics-only amortized forecaster until an own-history sample sufficient to initialize the per-name scale filter accrues; then attach this residual-hybrid pipeline. Total marginal cost per name per day: one filter update and one model evaluation.

Ex ante use of the score. Because the frontier score is a trailing window on GARCH-standardized residuals, it is available *ex ante*: a desk computes today's score from today's data and acts on it for tomorrow's risk number. What the evidence does not support is day-level model-switching. As Section 4.3 shows, the per-day oracle gap is unreachable (predicted-edge correlation +0.16), because which day a tail event lands is not forecastable from prior-day state, and a learned gate collapses to always-nonparametric. The edge is, in effect, negligible over the calm ∼90% of asset-days and concentrated (+2.7% pinball, DM 6.5) in the top misspecification decile and the most turbulent volatility regimes (an ∼fourfold gap, Section 4.3). The gating experiment provides no evidence that day-level switching improves on continuous use of the flexible model: on calm days the nonparametric/residual-hybrid family is statistically indistinguishable from the parametric benchmark (noninferior within the −0.25% margin where tested) and it has the advantage in stress, and the score serves as a monitor rather than a switch. The one setting in which the score does drive a switch is the per-name/standalone model, where nonparametric quantiles underperform on calm days; there the score gates, and its short detection latency (median zero, mean two to four days) costs little because the edge is back-loaded within a residual-shape stress episode.

Regulatory capital. The hybrid engine is an internal-models VaR/ES candidate that tops the accuracy table while passing the aggregate date-clustered exception test at both levels (per-name diagnostics show residual cross-sectional heterogeneity). This paper makes no capital-release claim: the predicted-vs-realized ES wedge conditions on each model's own breach set and doubles under the exact tail integral, so no defensible arithmetic converts it to capital. The narrower implication is that the ES-based component of an IMA charge scales with the ES forecast, but actual FRTB capital also involves liquidity-horizon adjustment, stressed-period

ES on a reduced factor set, constrained aggregation, non-modellable risk factors, and desk-level eligibility through backtesting and P&L attribution, so no one-for-one mapping from a single-name ES forecast to desk capital exists, and none is claimed. Establishing genuine calibration differences requires a jointly valid standard, and we therefore use the FZ0 comparison of Section 5 for the formal VaR/ES ranking. The ten-day rerun on 2000–2013 shows $\sqrt{h}$-scaled GARCH almost exactly calibrated through the crisis era while the direct-learned tail under-states it: multi-day capital economics would be regime-dependent in the direction that matters most.

A misspecification monitor. The score is a live, per-asset monitor: an asset entering its top decile sits where the parametric model's conditional loss runs $\sim$2–3% higher, and the input is a trailing window on GARCH residuals, so surveillance is real-time and cheap.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant in this paper that can produce a conditional risk forecast for an asset with *no return history*: an IPO or newly listed ETF gets day-one quantiles from its characteristics and its first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names. (The day-one forecaster is the raw amortized model; the residual-hybrid estimator of Section 2.2 attaches once a scale filter is estimable.) The practical numbers: the amortized forecast beats own-history benchmarks at every listing age, largest when young ($\sim$6–10% of pinball in the first trading month, full-scale panel), and below our 750-observation fitting floor a per-name GARCH is not reliably estimable in this panel. Section 2.3's ablation traces this as history accrues: characteristics carry the forecast at birth, and the name's own realized volatility takes over as it ages.

Trading implications. The universes where the nonparametric tail wins (crisis FX, volatility indices, credit spreads, frontier equities) are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physical-measure tail forecast does not by itself make bought crash insurance profitable, because the crash risk premium can survive conditioning on a correct forecast. The deployable side is the other one, *warehousing* (selling) the premium, with the model supplying tail control rather than alpha on the mean. The gating experiment provides no evidence that the score times individual crash days, so its role in a premium-harvesting book is continuous tail-risk measurement, not a market-timing signal.

Scenario generation. Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

7 Conclusion

A fixed-shape parametric filter is hard to beat where its shape assumption holds, and for most assets on most days it holds well. The contribution of this paper is to make the exception measurable. We identify a state, indexed by a real-time score of residual-shape stress, in which a distribution-free family has a large and significant advantage over the parametric benchmark, and we build an engine that captures that advantage while matching the benchmark elsewhere. This engine improves on the market-risk models banks commonly run, judged by FRTB-aligned diagnostics. We do not claim the full internal-models-approval machinery.

Three qualifications bound these conclusions. First, the score is predictive rather than a necessary or sufficient condition for the flexible model to win. High residual kurtosis coincides with ties in some markets and with a nonparametric win at moderate kurtosis in others, and in credit and freight the accuracy edges sit alongside calibration failures that the current tail stages leave open; the score orders the edge as an empirical regularity, and the data do not identify residual misspecification as its structural cause. Second, the study is daily-frequency, and an intraday extension that exploits high-frequency data is the natural next step. Third, the score is a monitor of the frontier rather than a rule for switching models from one day to the next. The day-ahead gap between the oracle and the achievable forecast is largely unforecastable noise, as Section 4.3 shows, so the score earns its keep by stratifying where the flexible model helps, not by timing a daily switch.

Acknowledgments

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Data and code availability

A public replication package reproducing every WRDS-based table and figure in this paper (the Bloomberg-based exhibits are preserved as-run with their derived series, terminal access having ended in mid-2026), estimation code for all stages of the engine ((2)–(5)), the misspecification score (9), the FRTB battery, the generative-posterior and SBC routines, and the exact walk-forward, seeds, and hyperparameter grids referenced in the text, is public at `github.com/sean-qin-usa/downside-risk-paper` (the repository README maps each table to its canonical script and result artifact; the strict-calibration audit is `code/job_fz_strict_calibration.py` → `results/fz_strict_calibration_results.json`); a citable archival snapshot (Zenodo DOI) will be minted for the accepted version. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author's institutional licenses and cannot be redistributed; the package includes the code to rebuild every panel from those sources, the derived non-proprietary series, and a synthetic example dataset (`toy_example.py`) that runs the full pipeline end-to-end without licensed data. The holdout's frozen specification, including its written-in-advance predictions, is the header of `code/job_wrds_holdout.py`, line-for-line diffable against the design-era pipeline in the same folder. Two numerical corrections are canonical in the package: the GJR-skew-t inverse CDF (history preserved in `code/frtb_bench.py`) and the exact-tail-integral ES. Table 6 is generated by `code/frtb_table.py` → `results/frtb_table_results.json`; the ten-day figures by `code/frtb_stress_exact.py` → `results/stress_es_results.json`; pre-correction artifacts are preserved in the repository history, not in the working tree. The analyses deferred in earlier drafts and since completed ship with their result files, and the remaining open items (Section 7) are tracked in the repository.

References

August A. Balkema and Laurens de Haan. Residual life time at great age. *Annals of Probability*, 2(5):792–804, 1974.

Giovanni Barone-Adesi, Kostas Giannopoulos, and Les Vosper. VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5):583–602, 1999.

Basel Committee on Banking Supervision. Minimum capital requirements for market risk. Technical report, Bank for International Settlements, 2019.

Pier Giovanni Bissiri, Chris C. Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130, 2016.

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

Victor Chernozhukov, Iván Fernández-Val, and Alfred Galichon. Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125, 2010.

Victor Chernozhukov, Kaspar Wüthrich, and Yinchu Zhu. Exact and robust conformal inference methods for predictive machine learning with dependent data. In *Proceedings of the 31st Conference on Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 732–749, 2018.

Peter F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4): 841–862, 1998.

Will Dabney, Georg Ostrovski, David Silver, and Rémi Munos. Implicit quantile networks for distributional reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1096–1105, 2018.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Springer, 1997.

Robert F. Engle and Simone Manganelli. CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381, 2004.

Tobias Fissler and Johanna F. Ziegel. Higher order elicitability and Osband's principle. *Annals of Statistics*, 44(4):1680–1707, 2016.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 1660–1672, 2021.

Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801, 1993.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011.

Peter Reinhard Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.

Wenxin Jiang and Martin A. Tanner. Gibbs posterior for variable selection in high-dimensional classification and data mining. *Annals of Statistics*, 36(5):2207–2231, 2008.

Roger Koenker and Gilbert Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

Paul H. Kupiec. Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84, 1995.

Alexander J. McNeil and Rüdiger Frey. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4): 271–300, 2000.

Maria Nareklishvili, Nicholas G. Polson, and Vadim Sokolov. Generative quantile Bayesian prediction. *arXiv preprint arXiv:2510.21784*, 2025.

Andrew J. Patton, Johanna F. Ziegel, and Rui Chen. Dynamic semiparametric models for expected shortfall (and Value-at-Risk). *Journal of Econometrics*, 211(2):388–413, 2019.

James Pickands. Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1): 119–131, 1975.

Nicholas G. Polson and Vadim Sokolov. Generative AI for Bayesian computation. *arXiv preprint arXiv:2305.14972*, 2023.

Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 3538–3548, 2019.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Chao Zhang, Yihuang Zhang, Mihai Cucuringu, and Zhongmin Qian. Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22(2): 492–530, 2024.

Gilles O. Zumbach. The RiskMetrics 2006 methodology. Technical report, RiskMetrics Group, 2007.

A Proofs

The four formal results of Section 2.4 are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]}(q-z)\,dF(z) \;+\; \tau\int_{(q,\infty)}(z-q)\,dF(z)$$

is finite for every q, and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed z, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q') - \rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau, 1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z < q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z < q\} - \tau = F(q^-) - \tau,$$

which equals $F(q) - \tau$ for all but countably many q. Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\,du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity*. For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\,du = \int_{q}^{q_F} \big(\tau - F(u)\big)\,du \;\geq\; 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q. Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q, then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$, and integrating over the interval yields $\frac{m}{2}(q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q - q_F)^2$. $\square$

Proof of Proposition 1 (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k-th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \ldots, e_n\}$ exists; when $k = n + 1$ (i.e. τ within $1/(n + 1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n + 1$ errors $e_1, \ldots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$) are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n + 1$ values is uniformly distributed on $\{1, \ldots, n + 1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\left(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\right) = \Pr(R \leq k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \leq \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n + 1$ gives $\tau \leq k/(n + 1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1}, because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. $\square$

Proof of Proposition 2 (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu - 2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu - 2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1 + o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}}\,(1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)}(1 + o(1)) \;\longrightarrow\; \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes existence of τ^*; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in Remark 1.) Under local truth t_{ν_1}, applying (8) with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. □

On the score-ordering: why it is an empirical regularity, not a theorem

We state the score-ordering as an empirical regularity rather than a theorem, and record here why no clean monotonicity theorem holds at the deployable levels. The tail-wedge result (Proposition 2) orders a given pair of Student-t tail indices only below a pair-specific crossover level $\tau^*(\nu_1, \nu_2)$; it does not deliver monotonicity of the wedge in the tail index at a fixed operational τ. More fundamentally, the innovations are unit-variance by construction, and for the unit-variance t_ν the level-τ quantile is $Q_\nu(\tau) = \sqrt{(\nu-2)/\nu}\, t_\nu^{-1}(\tau)$, which is *not* monotone in ν: as $\nu \downarrow 2$ the normalization $\sqrt{(\nu-2)/\nu} \to 0$ pulls the fixed-τ quantile back toward zero, so an arbitrarily heavy-tailed unit-variance t eventually has a fixed-τ quantile arbitrarily close to the center. The non-monotonicity is already active at the regulatory levels: at $\tau = 0.025$, $Q_5(0.025) \approx -1.99$ is *less* extreme than $Q_8(0.025) \approx -2.00$. A clean ordering theorem would therefore have to restrict the tail index to an explicit interval, isolate a range of τ on which $\partial Q_\nu(\tau)/\partial\nu$ has one sign, and control the density factor in the regret expansion; we do not pursue that narrow statement. The exact content we retain is the regret identity (Lemma 1) and the tail-wedge ordering (Proposition 2), and the frontier itself is documented empirically in Sections 4–5. □

Proof of Lemma 2 of the paper (generative sampler)

Let $\widetilde{q}(\cdot \mid s)$ be nondecreasing and left-continuous on $(0,1)$ and $\tau \sim U(0,1)$, and set $Z^* = \widetilde{q}(\tau \mid s)$ and $G(x) = \sup\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ (with $\sup \emptyset = 0$). Monotonicity makes $\{t : \widetilde{q}(t \mid s) \le x\}$ an interval with left endpoint 0, and left-continuity ensures the supremum is attained when the set is nonempty, so the set equals $(0, G(x)]$; hence $\{Z^* \le x\} = \{\tau \le G(x)\}$ and $\Pr(Z^* \le x) = G(x)$, i.e. Z^* has CDF G. Because G so defined is the upper generalized inverse of $\widetilde{q}$, and a left-continuous nondecreasing $\widetilde{q}$ is in turn the lower generalized inverse of G, the quantile function of Z^* is exactly $\widetilde{q}(\cdot \mid s)$. When $\widetilde{q}$ is an estimated curve, the argument applies verbatim to the estimate: the sampler is exact for the *fitted* conditional law, which is the claim. □